

Secure C# MCP Agent Demo

Professional architecture overview - Web UI, Native WPF Desktop, LLM agent, secured MCP tool server

Architecture Document

Version 1.4 | June 2026 | Demo / educational

Design Goals

Demonstrable release review UI
LLM-assisted parsing and synthesis
MCP server remains source of truth
Authorization is visible and testable

Security Boundary

The model cannot approve a release
approve_release requires
mcp:tools:release
All tool calls are auditable

Demo Modes

AnalyzerService: ready
PaymentGateway: blocked by CVE
Privileged mode: approval ticket

Native Desktop Demo

WPF C# E2E client
Launches MCP server
Live traces + logs
Manual log export

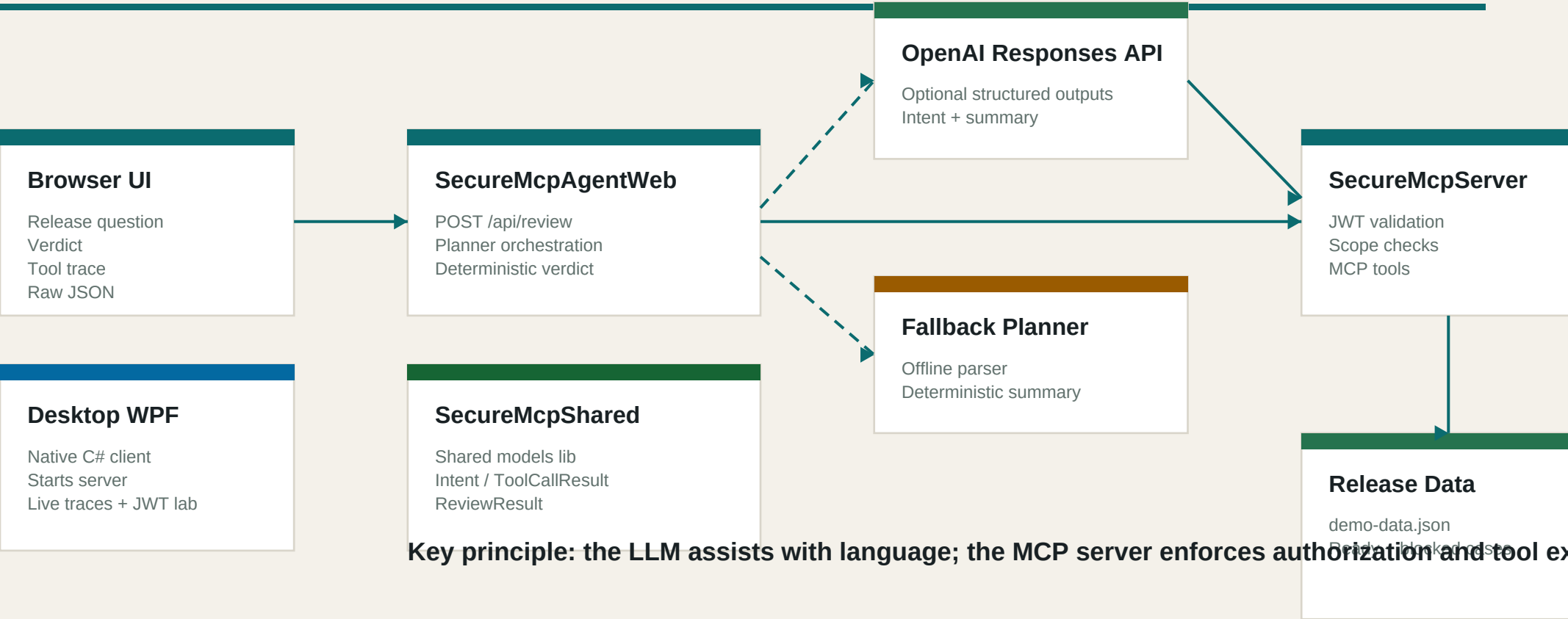
Shared Models

SecureMcpShared lib
ReleaseIntent / ToolCallResult
ReviewResult / TokenResponse
PaymentGateway / Desktop

The web agent may call OpenAI for structured intent parsing and final narrative, but every verdict is grounded in MCP tool results.

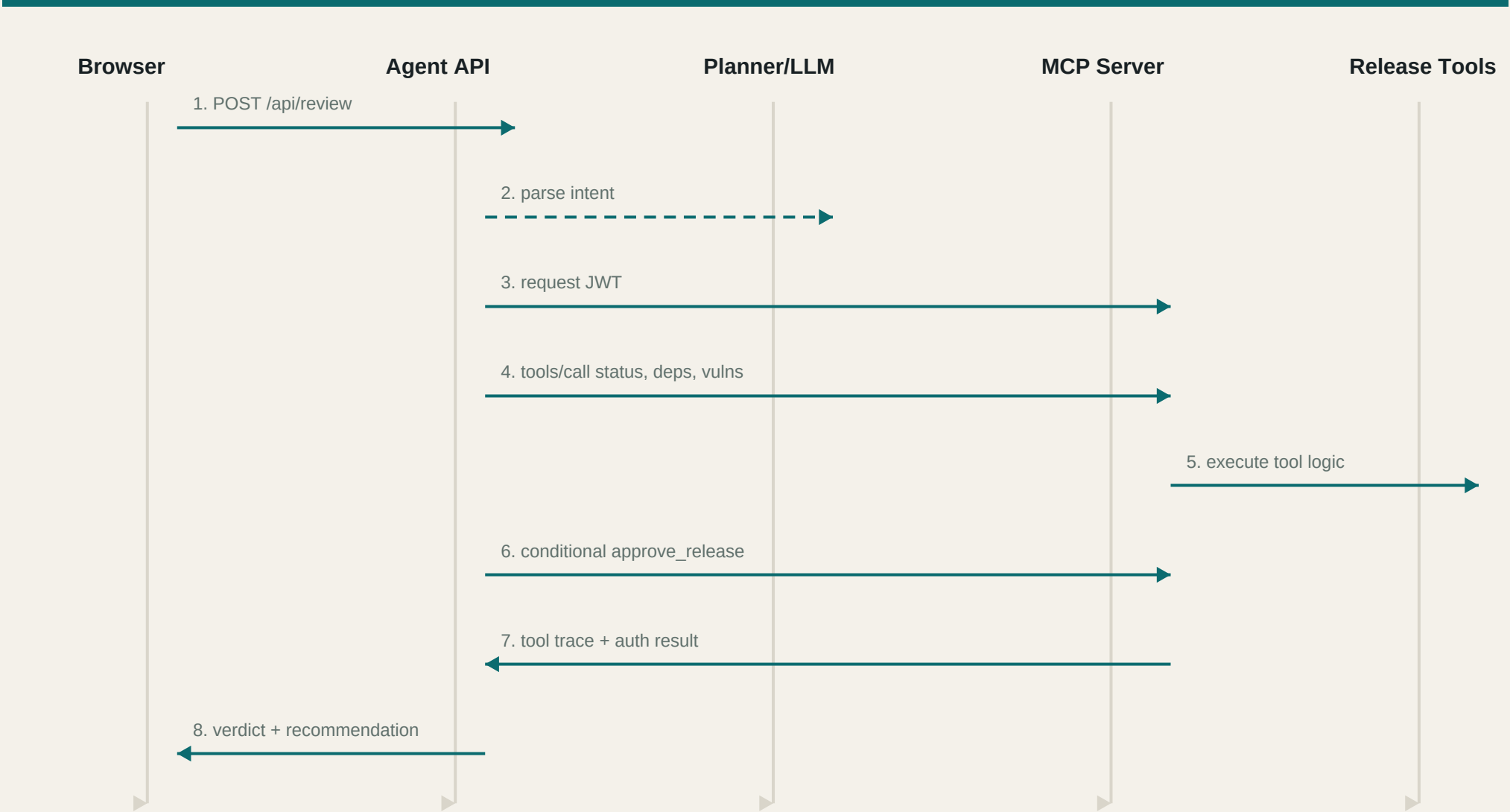
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System overview



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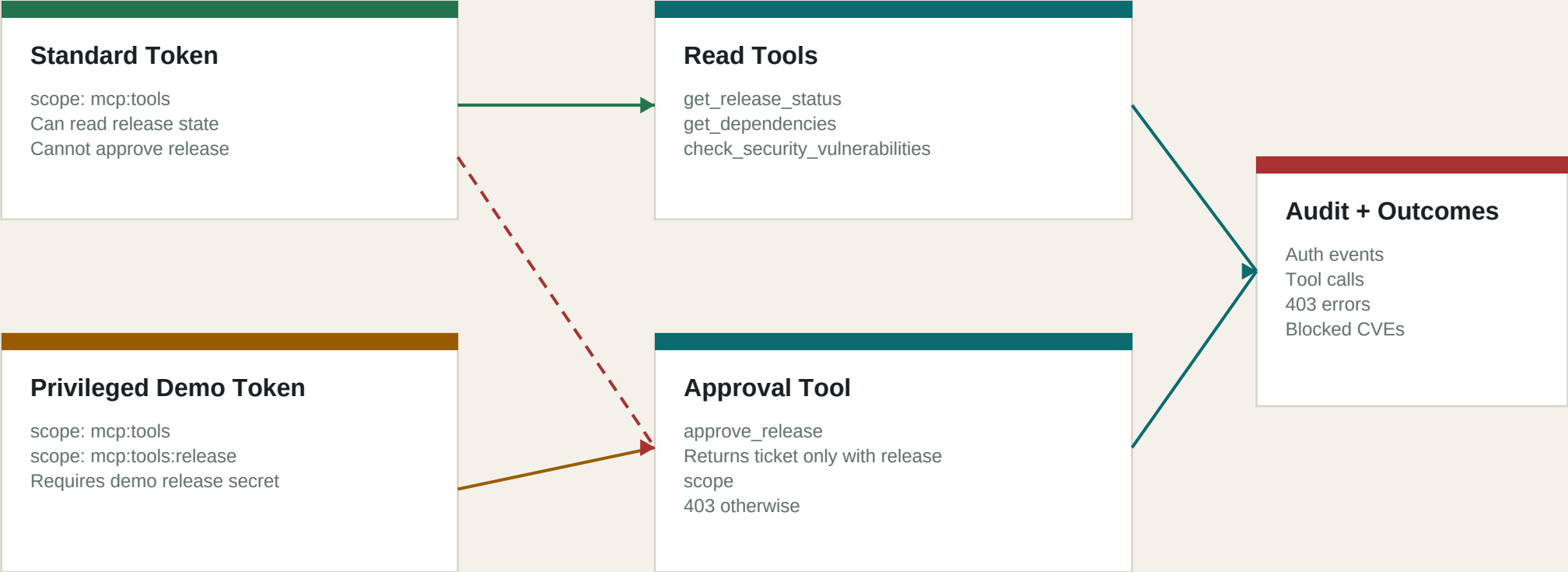
Sequence diagram - Release review



Approval is attempted only when release status is ready and no vulnerability findings are returned.

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Authorization boundaries



This boundary is intentionally visible in the UI: standard mode produces Needs Approval Scope; privileged mode can produce Approved.

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Runbook and hardening

Run Locally

1. Start SecureMcpServer
2. Start SecureMcpAgentWeb
3. Open <http://localhost:5080>
4. Try ready, blocked, approve

Configuration

OPENAI_API_KEY optional
OPENAI_MODEL default gpt-4o-mini
MCP_SERVER_URL
DEMO_RELEASE_CLIENT_SECRET

Validation

dotnet restore
dotnet build -c Release
dotnet test -c Release
Browser smoke checks

Production hardening

- Replace demo token endpoint with OAuth2/OIDC.
- Use asymmetric JWT signing and key rotation.
- Move secrets to managed secret storage.
- Send audit logs to an immutable sink.
- Add request limits, rate limiting, and full MCP transport tests.